Astro210 HW 13

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December 10, 2025

13.1 Cosmology missing steps

a)

Given ...

$$E(t) = \frac{4\sigma_{SB}}{c}\left(T^4 V\right), \quad P(t) = \frac{4\sigma_{SB}}{3c}T^4$$

We find ...

$$\frac{dE}{dt}\frac{4\sigma_{SB}}{c}\left(T^4 V\right) = \frac{4\sigma_{SB}}{c}\left(4T^3 V\frac{dT}{dt} + T^4\frac{dV}{dt}\right)$$

$$\boxed{\frac{4\sigma_{SB}}{c}\left(4T^3 V\frac{dT}{dt} + T^4\frac{dV}{dt}\right) = -\frac{4\sigma_{SB}}{3c}T^4\frac{dV}{dt}}$$

b)

Given ...

$$\frac{4\sigma_{SB}}{c}\left(4T^3 V\frac{dT}{dt} + T^4\frac{dV}{dt}\right) = -\frac{4\sigma_{SB}}{3c}T^4\frac{dV}{dt}$$

We find ...

$$\cancel{\frac{4\sigma_{SB}}{c}}\left(4T^3 V\frac{dT}{dt} + T^4\frac{dV}{dt}\right) = -\cancel{\frac{4\sigma_{SB}}{c}}T^4\frac{1}{3}\frac{dV}{dt}$$

$$\left(4\frac{\cancel{T^3}}{\cancel{T^4}}V\frac{dT}{dt} + \cancel{T^4}\frac{dV}{dt}\right) = -\cancel{T^4}\frac{1}{3}\frac{dV}{dt}$$

$$4\frac{1}{T}V\frac{dT}{dt} = -\frac{1}{3}\frac{dV}{dt} - \frac{dV}{dt}$$

$$\cancel{4}\frac{1}{T}V\frac{dT}{dt} = -\frac{\cancel{4}}{3}\frac{dV}{dt}$$

$$\frac{1}{T}V\frac{dT}{dt} = -\frac{1}{3}\frac{dV}{dt}$$

$$\boxed{\frac{1}{T}\frac{dT}{dt} = -\frac{1}{3V}\frac{dV}{dt}}$$

c)

Given ...

$$\frac{1}{T}\frac{dT}{dt} = -\frac{1}{3V}\frac{dV}{dt}, \quad V(t) \propto a(t)^3$$

We find ...

$$\frac{dV}{dt} = 3a^2\frac{da}{dt}$$

$$\frac{1}{T}\frac{dT}{dt} = -\frac{1}{a^{\cancel{3}}}\cancel{a^2}\frac{da}{dt}$$

$$\boxed{\frac{1}{T}\frac{dT}{dt} = -\frac{1}{a}\frac{da}{dt}}$$

d)

Given ...

$$\frac{1}{T}\frac{dT}{dt} = -\frac{1}{a}\frac{da}{dt}$$

We find ...

$$\frac{1}{T} = \ln(T)dT, \quad \frac{1}{a} = \ln(a)da$$

$$\boxed{\frac{d}{dt}\ln(T) = \ln(a)\frac{d}{dt}}$$

e)

Given ...

$$\lambda = \frac{2900\mu\text{m}}{T}$$

$$T = \frac{2900\mu\text{m}}{\lambda}$$

We find ...

$$\frac{d}{dt}\ln\left(\frac{2900\mu\text{m}}{\lambda}\right) = \ln(a)\frac{d}{dt}$$

$$\frac{d}{dt}\ln(2900\mu\text{m}) - \ln(\lambda) = \ln(a)\frac{d}{dt}$$

$$\boxed{\ln(\lambda) \propto \ln(a)}$$

f)

Given ...

$$\frac{1}{2}\left(\frac{dr}{dt}\right)^3 = \frac{GM}{r(t)} + k$$

$$\left(\frac{\dot{a}}{a}^2\right) = \frac{8\pi G r_0^2}{3} P(t) + \frac{2\kappa}{r_0^2}\frac{1}{a(t)^2}$$

$$M - \frac{4\pi}{3} p(t) r(t)^3$$

$$\frac{4G\pi r(t)^2 p(t)}{3} + \kappa = 1/2 \frac{dr}{dt}^3$$

$$r(t) = a(t) r_0$$

$$\frac{1}{2} r_0^2 \dot{a}^2 = \frac{4}{3} G\pi (a(t) r_0)^2 + \kappa = \frac{1}{2}\left(\frac{dr}{dt}\right)^2$$

$$\boxed{\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3} p(t) + \frac{2k}{r_0^2}\frac{1}{a(t)^2}}$$

g)

Given ...

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3} p(t) + \frac{2k}{r_0^2}\frac{1}{a(t)^2}$$

$$\left(\frac{\dot{a}}{a}\right)^2 = H(t)^2$$

$$k = 0$$

We find ...

$$H(t)^2 = \frac{8\pi G}{3c^2} v_c$$

$$\boxed{\frac{3c^2 H(t)^2}{8G\pi} = v_c}$$

h)

Given ...

$$H(t)^2 = \frac{8\pi G}{3c^2}\left[\frac{u_{r,0}}{a(t)^4}\frac{u_{m,0}}{a(t)^4 + u_{\Lambda,0}}\right]$$

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_r}{a(t)^4} + \frac{\Omega_r}{a(t)^4} + \Omega_\Lambda$$

$$\frac{da}{dt} = H_0\left[\frac{\Omega_r}{a(t)^2} + \frac{\Omega_r}{a(t)} + \Omega_\Lambda a(t)^2\right]^{1/2}$$

i)

$$\dot{a} = H_0 \left[\frac{\Omega_r}{a(t)^2} + \frac{\Omega_r}{a(t)} + \Omega_\Lambda a(t)^2 \right]^{1/2}$$

$$\frac{d}{dt}\dot{a} = H_0 \left[\frac{\Omega_r}{a(t)^2} + \frac{\Omega_r}{a(t)} + \Omega_\Lambda a(t)^2 \right]^{-1/2} \times \left[-2\frac{\Omega_r}{a(t)^3} - \frac{\Omega_r}{a(t)^2} + 2\Omega_\Lambda a(t) \right]$$

$$\ddot{a} = H_0 \frac{\left[-2\frac{\Omega_r}{a(t)^3} - \frac{\Omega_r}{a(t)^2} + 2\Omega_\Lambda a(t) \right]}{\left[\frac{\Omega_r}{a(t)^2} + \frac{\Omega_r}{a(t)} + \Omega_\Lambda a(t)^2 \right]^{1/2}}$$

$$\ddot{a} = H_0 \frac{\left[-2\frac{\Omega_r}{a(t)^3} - \frac{\Omega_r}{a(t)^2} + 2\Omega_\Lambda a(t) \right]}{\cancel{\left[\frac{\Omega_r}{a(t)^2} + \frac{\Omega_r}{a(t)} + \Omega_\Lambda a(t)^2 \right]^{1/2}}}$$

$$\boxed{\ddot{a} = H_0 \left[-2\frac{\Omega_r}{a(t)^3} - \frac{\Omega_r}{a(t)^2} + 2\Omega_\Lambda a(t) \right]^{1/2}}$$

13.2 Precision cosmological measurements

a)

Given . . .

$$H_{0_1} = 67 \text{kms}^{-1}\text{Mpc}^{-1}, \quad H_{0_2} = 74 \text{kms}^{-1}\text{Mpc}^{-1}$$

$$V_{rec} = d * H_0$$

We find . . .

$$\boxed{V_{rec_1} = 7.504 \times 10^6 \text{ m/s}, \quad V_{rec_2} = 8.288 \times 10^6 \text{ m/s}}$$

b)

Given . . .

$$c = \lambda\nu$$

$$\lambda = \frac{c}{\nu}$$

$$\nu = 22.23508 \text{ GHz}$$

We find . . .

$$\boxed{\lambda = 1.348 \times 10^{-2} \text{ m}}$$

c)

Given ...

$$V_{rec_1} = 7.504 \times 10^6\,\text{m/s}, \quad V_{rec_2} = 8.288 \times 10^6\,\text{m/s}$$
$$\lambda_{obsv} = \lambda_{rest}(1+z)$$
$$z = \frac{v}{c}$$

We find ...

$$z_1 = 2.503 \times 10^{-2}, \quad z_2 = 2.765 \times 10^{-2}$$
$$\nu_1 = 2.279 \times 10^{10}\,1/\text{s}, \quad \nu_2 = 2.285 \times 10^{10}\,1/\text{s}$$
$$\lambda_1 = 1.315 \times 10^{-2}\,\text{m}, \quad \lambda_2 = 1.312 \times 10^{-2}\,\text{m}$$
$$\boxed{\Delta\lambda = 3.347 \times 10^{-5}\,\text{m}}$$

13.3 Matter-dark energy equality

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a)

Given ...

$$\Omega_{\lambda_0} = (1+z_{eq})^3\Omega_{m_0}$$

We find ...

$$\frac{\Omega_{\lambda_0}}{\Omega_{m_0}} = (1+z_{eq})^3$$
$$\left(\frac{\Omega_{\lambda_0}}{\Omega_{m_0}}\right)^{1/3} - 1 = z_{eq}$$
$$\boxed{z_{eq} \approx 3.931 \times 10^{-1}}$$

b)

Given ...

$$\ddot{a} = H_0\left[\frac{\Omega_{r,0}}{a^2} + \frac{\Omega_{m,0}}{a} + \Omega_{\lambda,0}a^2\right]^{1/2}$$
$$1+z = \frac{1}{a} \Rightarrow a = \frac{1}{1+z}$$

We find...

$$\boxed{\ddot{a} = H_0(0.86734)}$$

c)

Given ...

$$\frac{H(t)^2}{H_0^2} = \frac{\Omega_{r,0}}{a(t)^4} + \frac{\Omega_{m,0}}{a(t)^3} + \Omega_\Lambda$$

We find ...

$$H(t) = H_0^2 \left[\frac{\Omega_{m,0}}{a(t)^3} + \Omega_\Lambda \right]^{1/2}$$

$$\boxed{H(t) = (H_0)^2 1.2083}$$

13.4 Curvature

Given ...

$$\alpha + \beta + \gamma = \pi + \frac{\kappa A}{r_{c,0}^2}, \quad \text{By spherical} \Rightarrow \kappa = 1$$

$$\alpha + \beta + \gamma = \theta_{tot}$$

$$A = \frac{\sqrt{3}}{4}\ell^2, \quad \ell = 1$$

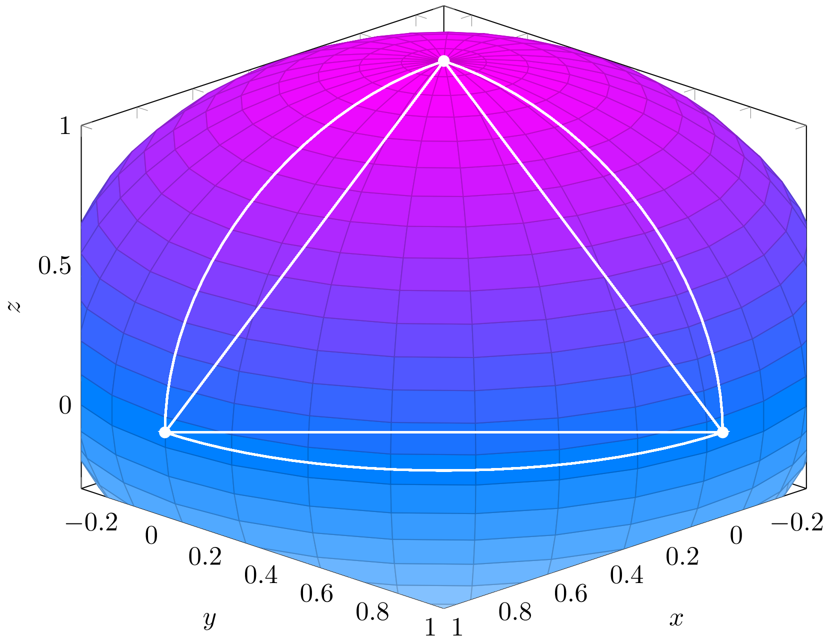
We find ...

$$\theta_{tot} = \pi + \frac{A}{r^2}$$

$$\boxed{\theta_{tot} = 3.141\,592\,664\,\text{rad}}$$

$$\boxed{\theta_{tot} - \pi = 1.066\,806 \times 10^{-8}\,\text{rad}}$$

Spherical triangle on a unit sphere



[1]

13.5 Proper distances

Given ...

$$r = \int_{t_{eq}}^{t_0} \frac{dt}{a(t)}$$

$$a(t) = \frac{t}{t_0}$$

We find ...

$$r = ct_0 \int_{t_{eq}}^{t_0} \frac{dt}{t}$$

$$r = ct_0 \ln(t)|_{t_{eq}}^{t_{t_0}}$$

$$r = ct_0(\ln(t_0) - \ln(t_{eq}))$$

$$r = ct_0 \left[\ln\left(\frac{t_0}{t_{eq}}\right)\right]$$

$$r = ct_0 \left[\ln\left(\frac{1}{a_{eq}}\right)\right]$$

$$r = ct_0 \left(\ln\left(1 + z\right)\right)$$

$$\boxed{r \approx 4.329 \times 10^{25}\,\mathrm{m} \approx 1.403 \times 10^{3}\,\mathrm{Mpc}}$$

[1] spent an hour procrastinating the actual homework making this